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THERMO-MECHANICAL VERIFICATION OF A FUEL PIN OF AN AP1000 REACTOR

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Introduction

Fuel rods employed in nuclear reactors are subjected to severe thermo-mechanical loading conditions during operation. The structural integrity of the fuel pin cladding is therefore a fundamental requirement to ensure safe and reliable reactor performance. In particular, the cladding must withstand the combined effects of internal pressure generated by fission gases, thermal gradients arising from reactor operation, and additional mechanical loads associated with the operating environment.

The present work focuses on the thermo-mechanical assessment of a nuclear fuel pin cladding under representative operating conditions. The analysis is primarily concerned with the evaluation of pressure-induced and thermally-induced stresses, while bending effects are neglected due to their comparatively limited contribution.

The study is conducted in accordance with the criteria prescribed by the ASME Boiler and Pressure Vessel Code (BPVC), Section III, which provides the allowable stress limits and structural verification procedures for nuclear components. The objective is to demonstrate that the stresses acting on the cladding remain below the admissible limits defined by the code, thereby ensuring the structural integrity of the fuel pin during operation.

To this end, both the mechanical and thermal stress components are evaluated at the inner and outer surfaces of the cladding. In particular, the hoop stress σ_h , radial stress σ_r , and axial stress σ_l are determined and subsequently combined to obtain the total stress state within the cladding material.

In addition to the structural verification, the sizing of the fuel pin gas plenum is addressed. The gas plenum plays a crucial role in accommodating the gaseous fission products released during reactor operation and in limiting the internal pressure increase within the fuel rod. Consequently, the required plenum volume and height are determined as part of the present analysis. Furthermore, the validity of the ideal gas assumption for the water vapor contained within the plenum is critically assessed.

The analysis is organized into several stages. First, a preliminary verification against cladding buckling is performed. Subsequently, the maximum internal pressure achievable within the fuel pin is estimated under conservative assumptions. Mechanical and thermal stresses are then evaluated, followed by the structural verification according to ASME BPVC Section III requirements. Finally, the gas plenum dimensions are determined in order to ensure adequate accommodation of fission gases during operation. [1]

Problem Statement

The AP1000 is the reference design. The goal of this assessment is the verification of the thermo-mechanical stresses of an AP1000 fuel pin cladding, and the sizing of the gas plenum to reduce the internal pressure down to acceptable values.

Data

Parameter	Symbol	Value
Thermal power	Q_{th}	3.4×10^9 W
Coolant pressure	$P_{coolant}$	1.551×10^7 Pa
Coolant inlet temperature	T_{in}	279.4 °C
Total coolant mass flow rate	$\dot{m}_{tot}$	1.43×10^4 kg s ⁻¹
Core coolant mass flow rate	$\dot{m}_{core}$	1.35×10^4 kg s ⁻¹
Bypass fraction	f_{bypass}	0.059
Fraction of heat deposited in fuel	f_{fuel}	0.974

Table 1: Main plant operational data adopted in the analysis. [2]

Parameter	Symbol	Value
Number of fuel assemblies	N_{ass}	157
Number of rods per assembly	$N_{\text{rod,ass}}$	264
Total number of fuel rods	N_{tot}	41448
Active fuel height	H	4.267 m
Lattice pitch	w	1.260×10^{-2} m
Cladding outer diameter	D_{co}	9.500×10^{-3} m
Cladding thickness	t_{clad}	5.715×10^{-4} m
Cladding inner diameter	D_{ci}	8.357×10^{-3} m
Fuel pellet diameter	D_f	8.192×10^{-3} m
Outer cladding radius	r_o	4.750×10^{-3} m
Inner cladding radius	r_i	4.179×10^{-3} m
Mean cladding radius	r_{avg}	4.465×10^{-3} m
Fuel cross-sectional area	A_f	5.271×10^{-5} m ²
Gas plenum cross-sectional area	A_{gas}	5.486×10^{-5} m ²

Table 2: Main fuel pin geometrical parameters adopted in the analysis. [2]

Assumptions

In order to simplify the thermo-mechanical analysis of the fuel pin, the following assumptions are adopted throughout the present work:

- The reactor is assumed to operate under steady-state nominal conditions;
- Only static loading conditions are considered in the structural assessment;
- Bending stresses, peak stresses, and local stress concentrations are neglected;
- The cladding material is assumed to be Zircaloy-4;
- The maximum internal pressure is evaluated assuming reactor depressurization conditions, namely:

$$p_o \approx 0$$

This value is subsequently adopted as the design internal pressure for the computation of the mechanical stresses acting on the cladding.

1 Preliminary Verification Against Buckling

A preliminary structural assessment of the fuel rods employed in the *AP1000* reactor is required in order to evaluate the stability of the cladding against buckling phenomena induced by the high pressure of the primary coolant system, namely $p_{\text{coolant}} = 15.5$ MPa.

Accordingly, starting from the mechanical properties of Zircaloy-4, which is adopted as the cladding material, it is necessary to determine the corresponding critical buckling pressure. Such a critical value should preferably exceed the coolant operating pressure, thereby ensuring the structural integrity and mechanical reliability of the fuel rods under normal operating conditions.

1.1 Zircaloy-4 Structural Data

The mechanical and elastic properties of the material, including Young's modulus, Poisson's ratio, yield strength, and thermal expansion coefficient, are strongly temperature dependent [3]. In order to ensure both physical accuracy and the required engineering conservatism in the thermo-mechanical assessment of the cladding, the elastic properties of Zircaloy-4, namely the Young's modulus E_{Zr} , Poisson's ratio ν_{Zr} , and the linear thermal expansion coefficient α_{Zr} , were evaluated by assuming as reference temperature the average radial cladding temperature, $T_{cl,ref}$, extracted at the axially most thermally stressed node.

The reference temperature was determined from the maximum inner and outer cladding temperatures previously calculated in the report entitled “*Thermal-Hydraulics in the Hot Subchannel of an AP1000 Reactor*”, namely $T_{ci} = 406.23^\circ\text{C}$ and $T_{co} = 347.17^\circ\text{C}$, respectively.

Accordingly, the adopted reference temperature is given by:

$$T_{cl,ref} = \frac{T_{ci} + T_{co}}{2} = \frac{(406.23 + 273.15) + (347.17 + 273.15)}{2} = 649.85 \text{ K}$$

Finally the mechanical and elastic properties of zircaloy-4 can be computed with the following relationships:

$$\alpha_{Zr} = \frac{6.72 \times 10^{-6} T_{cl,ref} - 2.07 \times 10^{-3}}{T_{cl,ref} - 308} = 6.72 \times 10^{-6} 1/K \quad (1)$$

$$E_{Zr} = (9.9 \times 10^3 - 5.669 (T_{cl,ref} - 273)) 9.81 \times 10^6 = 76.2 \text{ GPa} \quad (2)$$

$$\nu_{Zr} = 0.3303 + 8.376 \times 10^{-5} (T_{cl,ref} - 273) = 0.36 \quad (3)$$

Finally, the critical external pressure associated with the onset of elastic buckling of a Zircaloy-4 cladding tube can be evaluated according to the classical thin-shell formulation:

$$p_{cr} = \frac{E}{4(1 - \nu^2)} \left(\frac{t}{r_{avg}} \right)^3 = 45.97 \text{ MPa} \quad (4)$$

where E is the Young’s modulus of Zircaloy-4, ν is the Poisson’s ratio, t is the cladding thickness, and r_{avg} is the average cladding radius.

This expression provides the critical collapse pressure of the cylindrical shell under external loading and constitutes the reference criterion for verifying the structural stability of the cladding against coolant-pressure-induced buckling. Since $p_{cr} > p_{coolant}$, the structural stability of the cladding against collapse is ensured under nominal operating conditions. Consequently, the integrity of the fuel rod is maintained with respect to external pressure-induced buckling phenomena.

2 Calculation of the Maximum Theoretical Internal Pressure

In this phase, a conservative upper bound of the internal pressure is determined by assuming a fully depressurized reactor, i.e. $p_o \approx 0$.

For thin-walled cylindrical tubes ($t/r \ll 1$), the maximum theoretical internal pressure can be evaluated using Mariotte’s formula:

$$p_i^{max} = \frac{\sigma_y t}{r_{avg}} \quad (5)$$

where σ_h represents the allowable hoop stress of the cladding material, t is the wall thickness, and r_{avg} is the average radius of the tube. In this situation $p_i^{max} = 30.85 \text{ MPa}$.

3 Mechanical Stresses Calculation

The analysis of the mechanical stress state induced by pressure loading on the cladding was performed by abandoning the assumption of a depressurized reactor condition, adopted in the previous section for the calculation of maximum internal pressure. In order to accurately reproduce the actual operating conditions of the AP1000 reactor, the calculation was carried out by simultaneously considering both active pressure contributions:

- the internal pressure (p_i), assumed equal to the maximum theoretical gas pressure:

$$p_i = p_i^{max},$$

- the external coolant pressure (p_o), maintained at the nominal operating coolant pressure of the AP1000 primary circuit:

$$p_o = P_{coolant} = 15.5 \text{ MPa.}$$

Since the cladding can be modeled as a cylindrical component subjected to asymmetric radial pressure loading, the resulting stress state is inherently three-dimensional and was therefore evaluated through the rigorous application of Lamé's equations. This approach enables the determination of the three principal stress components — namely the hoop stress (σ_h), radial stress (σ_r), and longitudinal stress (σ_l) — across the entire cladding thickness.

In the present analysis, the exact mechanical stress values were computed at both the inner surface ($r = r_i$) and the outer surface ($r = r_o$) of the cladding.

At the inner surface ($r = r_i$), the mechanical hoop and radial stresses are given by:

$$\sigma_{h,m}(r_i) = \frac{p_i(r_i^2 + r_o^2) - 2p_o r_o^2}{r_o^2 - r_i^2} \quad (6)$$

$$\sigma_{r,m}(r_i) = -p_i \quad (7)$$

At the outer surface ($r = r_o$), the stress components are expressed as:

$$\sigma_{h,m}(r_o) = \frac{2p_i r_i^2 - p_o(r_i^2 + r_o^2)}{r_o^2 - r_i^2} \quad (8)$$

$$\sigma_{r,m}(r_o) = -p_o \quad (9)$$

Assuming a closed-end tube configuration for the cladding, the axial stress originates from the global force equilibrium acting on the end caps of the fuel rod. Under this assumption, the longitudinal stress is uniformly distributed throughout the wall thickness and can be evaluated as:

$$\sigma_{l,m} = \frac{p_i r_i^2 - p_o r_o^2}{r_o^2 - r_i^2} \quad (10)$$

The results are reported in the table below: The obtained results, particularly those concerning the radial

	σ_h [MPa]	σ_r [MPa]	σ_l [MPa]
Inner Wall ($r = r_i$)	104,8	-30,85	36,97
Outer Wall ($r = r_o$)	89,46	-15,5	36,97

Table 3: Computed mechanical stresses on the inner and outer wall of the cladding

stress values σ_r , are consistent with theoretical expectations. Specifically, at the inner surface σ_r coincides exactly with the internal pressure p_i , whereas at the outer surface it corresponds to the coolant pressure.

4 Thermal Stresses Calculation

The presence of a high linear heat generation rate induced by fission reactions within the fuel results in a pronounced radial temperature gradient across the cladding thickness. Based on the thermal profiles extracted at the axial hot spot in Assignment 2, the temperature difference between the inner and outer cladding surfaces is given by:

$$\Delta T_{clad} = T_{ci} - T_{co} = 59.06^\circ\text{C} \quad (11)$$

Owing to the higher temperature of the inner surface relative to the outer surface, the inner region undergoes greater thermal expansion. However, this differential expansion is constrained by the structural continuity of the cladding and the presence of the cooler outer region, leading to the development of a self-equilibrated thermoelastic stress field (secondary stress state). This stress distribution is obtained by applying the analytical thermoelastic solution for thick-walled cylinders subjected to a logarithmic radial temperature field.

Inner surface ($r = r_i$):

At the inner boundary, the material is at the highest temperature and is mechanically constrained by the cooler surrounding material. Consequently, the resulting hoop and axial stress components are compressive, while the radial stress vanishes due to the traction-free boundary condition:

$$\sigma_{h,th}(r_i) = \sigma_{l,th}(r_i) = \frac{\alpha_{Zr} E_{Zr} \Delta T_{clad}}{2(1 - \nu_{Zr}) \ln\left(\frac{r_o}{r_i}\right)} \left[1 - \frac{2r_o^2}{r_o^2 - r_i^2} \ln\left(\frac{r_o}{r_i}\right) \right] \quad (12)$$

$$\sigma_{r,th}(r_i) = 0 \quad (13)$$

Outer surface ($r = r_o$):

At the outer boundary, the lower temperature results in tensile stresses induced by the expansion of the hotter inner material. Accordingly, both hoop and axial stress components are tensile, while the radial stress remains zero:

$$\sigma_{h,th}(r_o) = \sigma_{l,th}(r_o) = \frac{\alpha_{Zr} E_{Zr} \Delta T_{clad}}{2(1 - \nu_{Zr}) \ln\left(\frac{r_o}{r_i}\right)} \left[1 - \frac{2r_i^2}{r_o^2 - r_i^2} \ln\left(\frac{r_o}{r_i}\right) \right] \quad (14)$$

$$\sigma_{r,th}(r_o) = 0 \quad (15)$$

The exact cylindrical thermoelastic solution further demonstrates that, under identical geometric and thermal conditions, the hoop and axial thermal stresses are identical:

$$\sigma_{h,th} = \sigma_{l,th} \quad (16)$$

In table 4 the values of thermal stresses are reported:

	$\sigma_{h,th}$ [MPa]	$\sigma_{r,th}$ [MPa]	$\sigma_{l,th}$ [MPa]
Inner Wall ($r = r_i$)	-24.69	0	-24.69
Outer Wall ($r = r_o$)	22.67	0	22.67

Table 4: Computed thermal stresses on the inner and outer wall of the cladding

5 Verification of Stresses with Tresca Criterion

The structural integrity of the AP1000 fuel rod cladding has been assessed in accordance with the nuclear design requirements of the ASME Boiler and Pressure Vessel Code (BPVC), Section III. The code requires the application of the Tresca yield criterion for the evaluation of the equivalent stress, i.e., the maximum absolute difference between the principal stresses:

$$\sigma_{equiv} = \max(|\sigma_h - \sigma_r|, |\sigma_r - \sigma_l|, |\sigma_l - \sigma_h|) \quad (17)$$

In accordance with the design guidelines, the verification has been performed separately for primary mechanical stresses and for combined loading conditions (primary + secondary thermal stresses), applying the corresponding allowable limits derived from the material properties of Zircaloy-4.

The design allowable stress is defined as:

$$S_m = \min\left(\frac{2}{3}\sigma_y, \frac{1}{3}\sigma_u\right) = 137.67 \text{ MPa} \quad (18)$$

where σ_y and σ_u denote the yield and ultimate tensile strength, respectively.

The allowable limit for primary plus secondary stresses is therefore:

$$3S_m = 3 \cdot \min\left(\frac{2}{3}\sigma_y, \frac{1}{3}\sigma_u\right) = 413.00 \text{ MPa} \quad (19)$$

With regard to mechanical stresses, the maximum value obtained using the Tresca yield criterion is

$$\sigma_a = 135.66 \text{ MPa} \quad (< S_m).$$

When primary stresses are combined with secondary stresses, the resulting equivalent stress is

$$\sigma'_a = 127.65 \text{ MPa} \quad (< 3S_m).$$

The design of the fuel rods employed in the AP1000 reactor fully satisfies the requirements imposed by the ASME Boiler and Pressure Vessel Code (BPVC). This outcome is not unexpected, given that the analysis concerns a reactor design that is already widely validated and deployed within the commercial pressurized water reactor (PWR) fleet.

6 Gas Plenum Sizing

The gas plenum sizing is performed under end-of-life fuel conditions, considering the most limiting scenario in which accumulated gaseous fission products and initially present impurities exert the maximum pressure on the cladding. The procedure strictly follows the sequential steps implemented in the computational script.

For the thermo-mechanical analysis, the following physical constants and design parameters are adopted:

- Universal gas constant: $R_{\text{gas}} = 8.3144 \text{ J}/(\text{mol} \cdot \text{K})$
- Avogadro's number: $N_A = 6.02214 \times 10^{23} \text{ mol}^{-1}$
- Energy released per fission: $E_f = 200 \text{ MeV} = 200 \cdot 1.60218 \times 10^{-13} \text{ J}$
- Design burn-up: $BU = 60000 \text{ MWd/tU} = 60 \text{ MWd/kgU}$
- Cumulative fission yield of Xe and Kr: $Y = 0.28$
- Gas release fraction from fuel: $R_r = 0.40$
- Impurity mass concentrations: $c_{\text{N}_2} = 25 \text{ ppm}$, $c_{\text{H}_2\text{O}} = 75 \text{ ppm}$
- Effective fuel density: $\rho_{\text{UO}_2} = 10400 \text{ kg/m}^3$

The total mass of uranium dioxide contained within the active fuel length H is computed from the pellet diameter d_p (equal to the fuel outer diameter D_f) as:

$$M_{\text{UO}_2} = \rho_{\text{UO}_2} \cdot \frac{\pi d_p^2}{4} \cdot H = 2.34 \text{ kg} \quad (20)$$

Impurities of nitrogen and water vapor trapped during fabrication are assumed to be fully released into the plenum due to the high operating temperatures. The corresponding number of moles is given by:

$$n_{\text{N}_2} = \frac{c_{\text{N}_2} M_{\text{UO}_2}}{u_{\text{N}_2}} = 0.0021 \text{ mol} \quad (21)$$

$$n_{\text{H}_2\text{O}} = \frac{c_{\text{H}_2\text{O}} M_{\text{UO}_2}}{u_{\text{H}_2\text{O}}} = 0.0097 \text{ mol} \quad (22)$$

To estimate irradiation-induced gas production, the mass of uranium contained in the fuel is first isolated:

$$M_{\text{U}} = M_{\text{UO}_2} \cdot \frac{u_{\text{U}}}{u_{\text{UO}_2}} = 2.06 \text{ kg} \quad (23)$$

The total number of fissions is then evaluated as:

$$N_f = \frac{BU_{\text{MWd/kg}} \cdot M_{\text{U}} \cdot 10^6 \cdot 24 \cdot 3600}{E_f} = 3.34 \times 10^{23} \quad (24)$$

The moles of xenon and krypton released into the plenum region are obtained as:

$$n_{\text{Xe-Kr}} = \frac{N_f \cdot Y \cdot R_r}{N_A} = 0.062 \text{ mol} \quad (25)$$

The total amount of gas accumulated at end of life is therefore:

$$n_{\text{tot,gas}} = n_{\text{Xe-Kr}} + n_{\text{N}_2} + n_{\text{H}_2\text{O}} = 0.0739 \text{ mol} \quad (26)$$

By imposing that the maximum internal pressure does not exceed the allowable cladding limit p_{max} (equal to the previously determined structural and buckling limit pressure $p_{i,\text{max}}$), the ideal gas law is applied at the plenum absolute temperature:

$$T_{\text{plenum}} = T_{ci} + 273.15 \quad (27)$$

The minimum required gas volume is:

$$V_{\text{min}} = \frac{n_{\text{tot,gas}} R_{\text{gas}} T_{\text{plenum}}}{p_{\text{max}}} = 13.52 \text{ cm}^3 \quad (28)$$

from which the minimum geometric height of the cylindrical plenum follows:

$$H_{\text{min}} = \frac{V_{\text{min}}}{\pi r_i^2} = 24.65 \text{ cm} \quad (29)$$

The final plenum height is then defined by adding the fixed allowance required for the spring spacer of the fuel column:

$$H_{\text{plenum,final}} = H_{\text{min}} + 15 \text{ cm} = 39.65 \text{ cm} \quad (30)$$

Finally, in order to assess the validity of the ideal gas assumption for the water vapor fraction, its partial pressure is evaluated as:

$$p_{\text{H}_2\text{O,ideal}} = \frac{n_{\text{H}_2\text{O}} R_{\text{gas}} T_{\text{plenum}}}{V_{\text{min}}} = 40.68 \text{ bar} \quad (31)$$

This value would typically be compared with the saturation pressure $p_{\text{H}_2\text{O,sat}}$ at temperature T_{ci} using XSteam [4]. However, in the implemented script this function returns **NaN**, which is physically justified. The cladding inner surface temperature T_{ci} under nominal AP1000 operating conditions exceeds the critical temperature of water:

$$T_{\text{crit}} = 373.946^\circ\text{C} = 647.096 \text{ K} \quad (32)$$

Above the critical point, water exists in a supercritical state where no liquid-vapor phase boundary exists and the concept of saturation pressure is no longer defined. Consequently, steam tables become inapplicable and the saturation function is undefined, leading correctly to a non-numerical output.

Since the operating temperature is significantly higher than the critical temperature and the calculated partial pressure $p_{\text{H}_2\text{O,ideal}}$ remains several orders of magnitude below the critical pressure of water (22.06 MPa), the vapor phase is highly rarefied and intermolecular interactions can be neglected. This confirms the physical consistency and validity of the ideal gas assumption adopted in the model.

Conclusions

The thermo-mechanical assessment of the AP1000 fuel pin cladding has been carried out in accordance with the ASME Boiler and Pressure Vessel Code (Section III), combining structural verification under mechanical and thermal loading with a conservative evaluation of internal gas pressure evolution over the fuel lifetime.

The preliminary buckling analysis demonstrated a comfortable safety margin against external pressure-induced instability, with a critical buckling pressure significantly higher than the nominal coolant pressure. This confirms that, under normal operating conditions, the cladding maintains structural stability against collapse phenomena.

The subsequent evaluation of internal pressure, based on a fully conservative depressurized scenario, provided an upper bound for the mechanical loading state of the cladding. Using this limit, the detailed stress analysis showed that both hoop and axial stresses remain within admissible values when compared to the allowable stress intensity defined by the ASME code. In particular, the Tresca equivalent stress satisfies both the primary and the combined (primary plus secondary) stress limits, confirming compliance with design requirements.

Thermal stresses, arising from the radial temperature gradient across the cladding wall, introduce a secondary self-equilibrated stress field. Although non-negligible, their magnitude remains limited compared to the primary mechanical stresses and does not compromise structural integrity. The combination of thermal and mechanical contributions still results in stress levels below the allowable thresholds, confirming the robustness of the design under realistic operating conditions.

The gas plenum sizing analysis further demonstrated that the required free volume at end-of-life conditions is driven primarily by the accumulation of fission gases (Xe and Kr), with additional minor contributions from fabrication impurities. The resulting plenum height ensures that the internal pressure remains bounded by the maximum allowable cladding pressure. The computed water vapor partial pressure, although significant, does not invalidate the ideal gas assumption due to the supercritical thermodynamic state of water at operating temperatures and the very low density regime considered.

Overall, the obtained results confirm that the AP1000 fuel rod cladding design satisfies all structural, mechanical, and thermo-mechanical constraints imposed by regulatory standards. The analysis highlights the inherent conservatism embedded in the design margins and supports the reliability and safety of the fuel pin configuration under both nominal and end-of-life operating conditions.

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